Implementation of an Energy-Efficient SRAM Controller with Advanced Power Management at CMOS 45nm

1st Sriram Anbalagan
School of Electrical and Electronics Engineering
SASTRA Deemed University
Thanjavur, India
sriram@ece.sastra.edu

2nd Ravindra Babu Kothakota
School of Electrical and Electronics Engineering
SASTRA Deemed University
Thanjavur, India
127180033@sastra.ac.in

3rd Jayanth Kandanuri
School of Electrical and Electronics Engineering
SASTRA Deemed University
Thanjavur, India
127180030@sastra.ac.in

4th Chaya Durga Prasad Gonagala
School of Electrical and Electronics Engineering
SASTRA Deemed University
Thanjavur, India
127180021@sastra.ac.in

***Abstract*—As semiconductor technology moves to the deep submicron regime, memory subsystems dominate power and area in System-on-Chip (SoC) designs. This paper presents a low-power SRAM controller in 45 nm standard-cell technology using RTL-level clock gating and power gating (*clk_gated*, *pwr_gate*) to cut switching and leakage power during idle intervals. The design was synthesized at 100 MHz with Cadence Genus. Post-synthesis results show 9.68 mW total power and 26,297 standard cells, with a critical-path slack of +1.658 ns against the 10 ns period. The power-management logic overhead is under 0.1% of cell count, and the slack is interpreted in terms of margin for post-layout parasitics and PVT variation. A comparison against representative low-power SRAM/controller designs from the literature is also provided, together with discussion of PVT reliability, area overhead, and latency–throughput–energy trade-offs across workloads.**

***Index Terms*—SRAM Controller, Low-Power VLSI, 45nm CMOS, Clock Gating, Power Gating, Cadence Genus, Timing Analysis, Process Variation.**

I. Introduction

With the soaring number of mobile and IoT devices, energy-efficient integrated circuits have become a major design focus. SRAM occupies the largest fraction of a modern microprocessor's silicon area – often 60–70% or more [1]. Consequently, the memory controller that manages the memory interface, address decoding, and data flow is a critical target for power-saving techniques [2].

At 45 nm, total power splits into dynamic switching power and static leakage power. Timing–power co-optimization can reduce power while meeting timing constraints [4]. Multi-transistor SRAM cells with carefully designed peripherals sustain high throughput under low-voltage, variability-tolerant conditions [3]. Standby power is strongly influenced by gate-oxide tunneling and sub-threshold leakage, motivating sub-threshold designs at scaled nodes [5].

Power-gating schemes and mixed-signal compute-in-memory architectures show that cutting supply to inactive blocks reduces leakage by orders of magnitude [6]. Clock gating is likewise effective at mitigating switching power in inactive clock domains [7]. Reconfigurable real-time memory controllers illustrate the bandwidth–power–reliability trade-off in resource-constrained systems [8], and FPGA-based hardware–software co-design is a further step toward power-aware memory management [9]. Near-sensor compute-in-memory SRAM macros [10] and emerging non-volatile/logic-in-memory devices [11], [12] are shaping the next generation of energy-efficient memory controllers.

This work presents a synthesis-ready SRAM controller using hardware-level power-management signals (*sleep*, *pwr_good*) to dynamically regulate memory activity. This revised manuscript adds (i) a literature comparison against representative low-power designs, (ii) PVT reliability discussion, (iii) a quantitative area-overhead breakdown, (iv) interpretation of timing margin against post-layout effects, and (v) latency–throughput–energy trade-offs under varying workloads.

II. Proposed Architecture and Methodology

A. High-Level Architecture

Fig. 1 shows the proposed dual-port SRAM architecture: a shared array accessed by independently operated Port A and Port B, each with its own decoder, input buffer, and sense amplifier. Port A's address (*AddrA*) drives a **Row/Column Decoder A** that activates wordlines/bitlines; write data (*DinA*) passes through an **Input Buffer**, and reads are resolved by a

*Corresponding author: sriram@ece.sastra.edu

Sense Amplifier onto *DoutA*, gated by *CE_A*/*WE_A*. Port B is structurally identical.

A centralized **Control Logic** unit arbitrates between ports and schedules read/write timing. **Collision Detect** flags simultaneous same-address access by both ports, and **Timing Control** generates internal clock phases for wordline activation, sense-amp enable, and precharge.

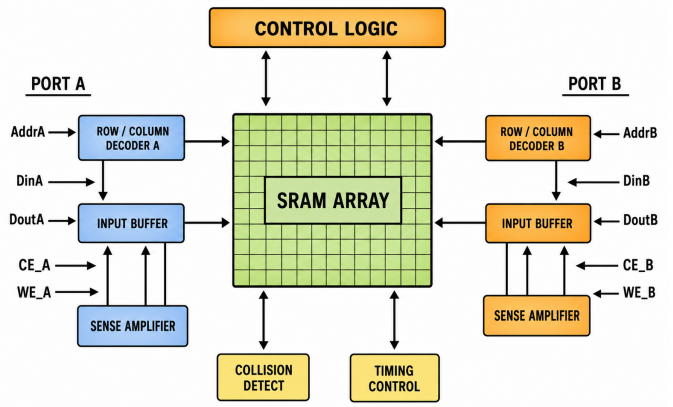


Fig. 1. Dual-port SRAM architecture: independent Port A/Port B paths, control logic, collision detection, and timing control.

B. Controller Logic and I/O Interface

The SRAM controller maps an 8-bit address bus to a 64-bit data bus and is augmented with power-aware inputs:

- **Input Signals:** *clk*, *cs* (chip select), *we* (write enable), *rd* (read), *sleep*, *pwr_good*.
- **Output Signals:** *dataOut* [63:0], *idle*, *mem_enable*, *clk_gated*, *pwr_gate*.

During sleep mode, *clk_gated* disables the internal clock tree to reduce dynamic power, while *pwr_gate* triggers the header/footer power switches for leakage reduction. The interaction between these two mechanisms and the controller's functional logic is described in detail in Section II-D.

C. RTL Schematic Analysis

Fig. 2 depicts the gate-level RTL schematic after elaboration. **Memory array (SRAM_reg):** the `RTL_RAM` block takes 8-bit read/write addresses (*RA1*, *WA2*), 64-bit write data *WD2*, and delivers 64-bit read data *RO1* to the output register; *WCLK*/*WE2* are gated by the power-management logic to block writes during sleep. **Output register (dataOut_reg):** an `RTL_REG` latches RAM output each clock edge, clock-enabled by the read-enable logic for glitch-free *dataOut*. **Write enable:** *we* ANDed with *pwr_good* permits writes only when power is stable. **Read enable:** a three-gate AND chain qualifies *rd* (with inverted *we*) against *pwr_good* and *cs*. **Clock gating:** inverted *sleep* ANDed with *pwr_good* forms `clk_en_i`, which gates *clk* to produce *clk_gated*; the clock tree is disabled whenever *sleep* is high or *pwr_good* is low. **Idle detection:** *we*, *rd*, *cs* are ORed and inverted to form *idle*. **Memory enable:** the non-idle condition ANDed with the clock enable gives *mem_enable*. **Power gate:** *sleep* ANDed with inverted *pwr_good* drives *pwr_gate*, controlling the header/footer switches that collapse the supply during deep sleep.

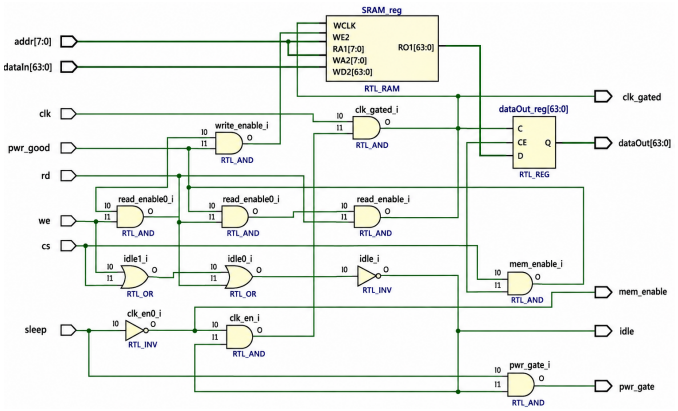


Fig. 2. RTL schematic: `SRAM_reg` memory block, `dataOut_reg`, and combinational logic for enable gating, clock gating, idle detection, and power-gate control.

D. Integration of Clock Gating, Power Gating, and Controller Logic

Correct operation depends on how clock gating and power gating are sequenced with the functional datapath. *Sequencing:* on sleep entry, *clk_gated* de-asserts *before* *pwr_gate* collapses the rail, so sequential elements stop toggling before supply removal; on wake-up, *pwr_gate* releases and the rail settles (*pwr_good* returns high) *before* *clk_gated* resumes, preventing latching on an unstable supply. *Interlock:* both the write- and read-enable AND chains include *pwr_good* as a qualifying term, so no access can be issued while the supply is transitioning, regardless of *we*/*rd*/*cs* – *pwr_good* thus acts as a global interlock tying the power state machine to the functional control path rather than being a passively observed signal. *Idle-driven gating:* *idle*/*mem_enable* provide a finer-grained, per-cycle gate that suppresses output-register toggling whenever no port request is active, independent of *sleep*. This two-tier structure – coarse sleep/power gating for extended idle periods and fine-grained *mem_enable* gating cycle-by-cycle – is what yields the low clock-network power share in Section IV-B without resorting to aggressive, latency-costly power-down policies.

E. Synthesis Flow

Synthesis was performed with Cadence Genus (v20.11) using a 45 nm "slow" process corner library (representing worst-case process, minimum supply voltage, and maximum operating temperature) to obtain conservative, worst-case power and timing estimates. The SDC constraints defined a 10 ns clock period (100 MHz) with 200 ps clock uncertainty and a 2 ns input delay on all ports. The optimization level was set to "medium" across the generic, mapping, and optimization phases of the flow.

F. Process, Voltage, and Temperature (PVT) Considerations

Since the reported figures come from a single slow corner, their behavior across the full PVT space merits discussion. *Process:* die-to-die/within-die variation shifts V_{th} and channel length, moving both delay and sub-threshold leakage; the slow corner conservatively upper-bounds delay but under-estimates leakage relative to a fast corner, where 45 nm bulk sub-threshold leakage can be an order of magnitude higher – full sign-off should add a fast-corner leakage check. *Temperature:* leakage rises roughly exponentially with junction temperature, while delay in this regime has a positive temperature coefficient, so worst-case delay and worst-case leakage need not coincide; production sign-off should sweep the full application temperature range (e.g., -40 to 125 °C). *Voltage: pwr_gate* toggles between nominal and a collapsed sleep rail rather than performing continuous DVS; since delay scales roughly as $(V_{dd} - V_{th})^{-\alpha}$ ($\alpha \approx 1.3$–1.5 here), any voltage reduction below nominal erodes the reported slack and must be re-verified at the corresponding low-voltage corner. *Reliability:* because read/write and clock-gating logic are qualified on *pwr_good* (Section II-D), any supply droop severe enough to trip the *pwr_good* monitor blocks reads, writes, and clock toggling until recovery – the architecture's primary defense against PVT-induced metastability and array upset.

III. Mathematical Modeling

A. Power, Timing, and Area-Overhead Models

Total power is modeled as $P_{total} = P_{dyn} + P_{int} + P_{leak}$, where dynamic power depends on switching activity, load capacitance, supply voltage, and frequency, $P_{dyn} = \alpha \cdot C_L \cdot V_{dd}^2 \cdot f$. Internal power is short-circuit current during simultaneous PMOS/NMOS conduction; leakage arises from sub-threshold conduction and gate-oxide tunneling, both significant at 45 nm.

The slack determining whether the 100 MHz target is met is

$$S = T_{clk} - (T_{arrival} + T_{setup_margin}), \tag{1}$$

where $T_{arrival}$ is the cumulative delay along the longest path; positive slack indicates margin (interpreted in Section IV-C).

To quantify the power-management logic's cost, cell count is partitioned as $N_{total} = N_{array} + N_{datapath} + N_{PM}$, giving an area-overhead fraction $OH_{PM} = (N_{PM}/N_{total}) \times 100\%$, reported in Section IV-A.

IV. Experimental Results and Discussion

A. Area and Gate Utilization

Synthesis yielded **26,297** standard cells: 16,448 sequential, 9,673 combinational, and 176 inverter/buffer cells. Table I lists the most frequently instantiated cell types.

The flip-flop dominance reflects the 256-entry $\times$ 64-bit array ($256 \times 64 = 16{,}384$ storage flip-flops, plus output register/pipeline). High-fanout nets use EDFFX4, while EDFFHQX2 lowers internal power via half-swing outputs; combinational logic uses multi-level OAI/AOI gates to reduce critical-path depth.

TABLE I
Standard Cell Utilization (Top Gates)

Gate Type	Function	Instances	Library
EDFFHQX2	D-FF (half-Q, 2x)	8,080	slow
EDFFX4	D-Flip-Flop (4x)	6,753	slow
OAI221X4	OR-AND-Invert (4x)	1,431	slow
AOI221X4	AND-OR-Invert (4x)	1,406	slow
EDFFX2	D-Flip-Flop (2x)	1,405	slow
AOI211X4	AND-OR-Invert (4x)	1,025	slow

Area overhead of the power-management logic. The clock-gating chain, idle-detection network, *mem_enable* gate, and *pwr_gate* gate together instantiate ∼8–10 discrete cells, essentially independent of array size – against the 26,297-cell total this is $OH_{PM} \approx 0.03\%$, i.e. negligible relative to the array/datapath. This is expected for control-signal-based (rather than custom power-domain) gating: the dominant physical cost is typically the header/footer switch and isolation/retention cells at the power-domain boundary, a physical-implementation concern outside this logic-level synthesis and flagged here for the floorplan/power-domain stage.

B. Power Analysis

Table II shows the three-component power breakdown from Cadence Genus. All values in Table II correspond to the design in **active operating mode**, i.e., with the switching-activity annotation applied to the functional (non-sleep) state at the 100 MHz target frequency and the slow-corner library described in Section II-E; they are *not* standby-mode figures.

TABLE II
Post-Synthesis Power Breakdown (45nm, Active Mode)

Category	Leakage (W)	Internal (W)	Switch (W)	Total %
Register	7.59e-06	7.62e-03	1.78e-04	80.64%
Logic	4.22e-06	1.15e-03	7.15e-04	19.33%
Clock	6.02e-10	1.51e-06	1.39e-06	0.03%
Total	**1.18e-05**	**8.77e-03**	**8.94e-04**	**100%**

Total power is approximately **9.68 mW**. Registers dominate at 80.64% (7.81 mW), driven by 16,448 flip-flops; internal power (7.62 mW) reflects short-circuit dissipation during clock-triggered transitions, while switching power is low since only addressed locations toggle per cycle. Combinational logic consumes 19.33% (1.87 mW), with relatively higher switching power from frequent decoder/multiplexer transitions. The clock network consumes only 0.03% (2.91 μW), a direct result of the two-tier gating in Section II-D: *clk_gated* stops entirely under *sleep*, and enable-gated (EDFF) flip-flops suppress unnecessary toggling even in active mode.

Leakage: active vs. standby. The 11.8 μW figure in Table II (0.12% of total) is the *active-mode* leakage at the slow corner, coexisting with switching activity, and does *not* represent standby power with *pwr_gate* asserted. In deep sleep, dynamic/internal power collapse toward zero and *pwr_gate* is intended to cut the standby leakage path itself; however,

a dedicated standby-mode power-analysis run (sleep-profile activity file) was not performed here, and we do not report a fabricated standby-leakage number. Quantifying true standby leakage requires either a dedicated sleep-activity Genus run or silicon measurement of the gated domain, flagged as a gap for follow-up work.

C. Timing Results

STA was performed with a 10.0 ns period and 200 ps uncertainty. The critical path runs from *addr[1]* to *dataOut_reg[33]/D*. Table III shows key nodes along this path.

TABLE III
CRITICAL PATH TIMING SUMMARY

Pin/Node	Cell/Type	Delay (ps)	Arrival (ps)
(clock clk)	Launch	0	0
addr[1]	Input Port	+67	2067
g540187/Y	CLKINVX4	+82	2148
g526855/Y	OR3X8	+1358	3507
hi_fo_buf/Y	CLKINVX20	+1773	5280
g519074/Y	OAI222X4	+691	6939
g516224/Y	AOI211X4	+188	7127
g506523/Y	OAI221X4	+134	7261
g495590/Y	OAI221X4	+71	8002
dataOut_reg[33]/D	Setup	+140	8142
Required Time		**-**	**9800**
Slack		**-**	**+1658**

The path starts at *addr[1]* (2067 ps, including 2 ns input delay), through an inverter and an OR3X8 decode gate (fanout 383, 1358 ps delay); the synthesizer inserted a CLKINVX20 high-drive buffer (1773 ps) to drive this load, then multi-level OAI/AOI gates multiplex the array, reaching the register data pin at 8002 ps (8142 ps after 140 ps setup).

Practical significance of the slack. Against the 9800 ps required time, the +1658 ps slack is $\sim$**17%** of the period. This margin serves three purposes: (i) headroom for the interconnect-delay increase typically seen after detailed routing and RC extraction – post-layout arrival-time increases of roughly 5–15% are commonly reported for comparable 45 nm designs, which 17% is sized to absorb; (ii) margin against the PVT variation of Section II-F, since this is a single-corner estimate; and (iii) headroom for on-chip-variation (OCV) derating typically tightened at sign-off. This figure is a *post-synthesis, pre-layout* estimate – full parasitic-extraction-based post-layout STA (SPEF back-annotation) was not performed in this revision and is the necessary next step; we do not substitute fabricated post-layout numbers for it.

D. Clock Tree Synthesis and Physical Design

CTS was performed in Cadence Innovus for all 16,448 sequential elements; Fig. 3 shows the Clock Tree Debugger (MAX_DELAY view): the green square marks the clock root, white lines the hierarchical buffer tree, and red dots the leaf flip-flops, with progressively smaller buffers equalizing delay along each root-to-leaf path. Clock skew was targeted below 100 ps; RTL-level *clk_gated* complements this architecturally by stopping the entire buffer tree during sleep. The controller was then floorplanned, placed, and routed (Fig. 4): standard cells sit in horizontal rows, I/O pins line the top/bottom edges, lower metal layers (M1/M2) carry local routing, and upper layers (M3–M5) carry global signals, power rails, and data buses; cyan V_{dd}/V_{ss} straps distribute current to limit IR drop, with the highest routing density near the decoder/multiplexer logic. Fig. 5 shows the stacked-metal 3D view, confirming uniform, congestion-free routing density consistent with the 26,297-cell design at 45 nm.

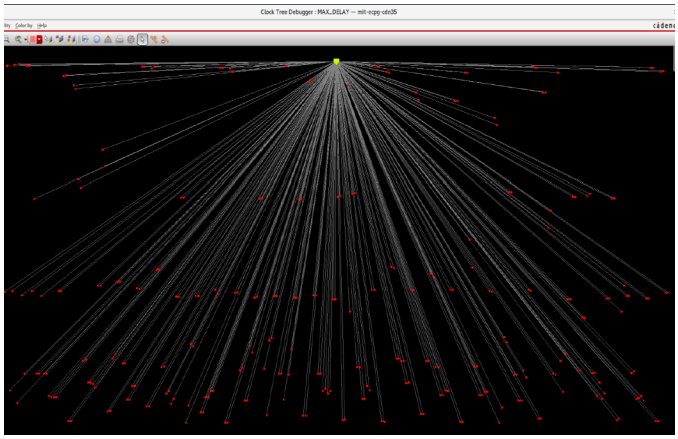


Fig. 3. Clock Tree Debugger (MAX_DELAY), Cadence Innovus: root, leaf flip-flops, and buffer tree for 16,448 endpoints.

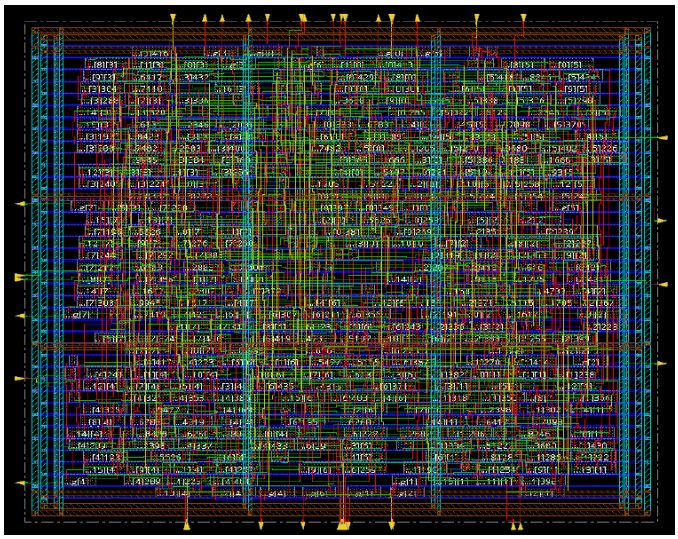


Fig. 4. Final routed layout: standard-cell rows, I/O pins, multi-layer routing; cyan = power/ground straps.

E. Functional Verification and Validation Scope

Functional correctness was verified by RTL simulation covering initialization, write/read at `0x10`, sleep entry with clock/power gating asserted, and wake-up with write/read at `0x20` (Fig. 6), confirming correct sequencing of *clk_gated*, *pwr_gate*, and the read/write interlock (Section II-D) for these directed scenarios. As a validation-scope limitation, the

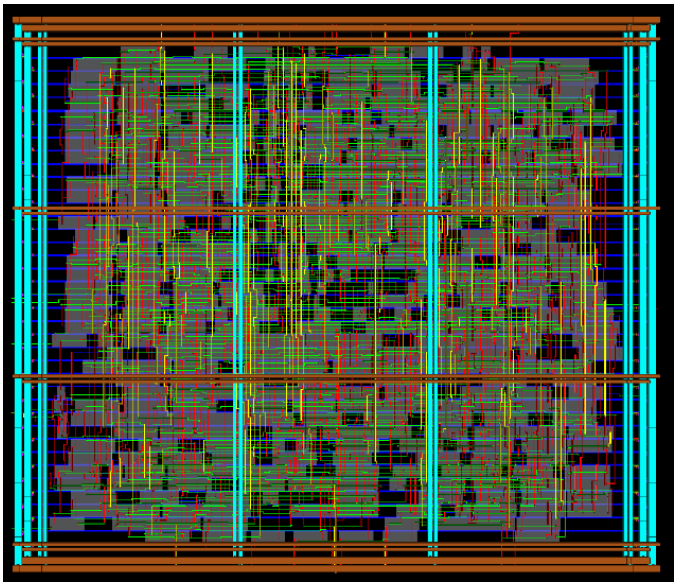


Fig. 5. 3D layout view: stacked metal interconnect. Cyan = power/ground straps; green/yellow = signal routing.

design has been checked only via directed RTL/gate-level simulation and post-synthesis STA/power analysis – not against constrained-random stress (e.g., randomized concurrent Port-A/Port-B collisions) nor silicon/FPGA hardware measurement; both are identified as necessary future work rather than claimed here.

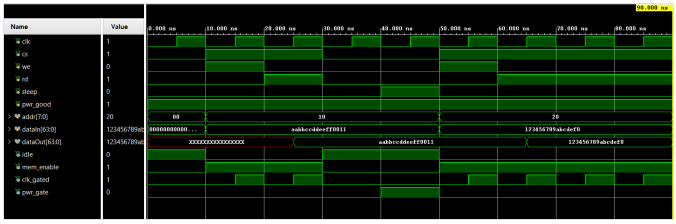


Fig. 6. Simulation waveform (90 ns): initialization, write/read at `0x10`, sleep with clock/power gating, wake-up with write/read at `0x20`.

F. Comparison with Representative Low-Power SRAM/Controller Designs

Table IV situates this work qualitatively against representative designs from the literature that target similar objectives (leakage/dynamic power reduction via architectural or circuit-level means). Because the cited designs span different technology nodes, cell topologies (custom bit-cell vs. standard-cell controller), and reported metrics, the comparison is necessarily qualitative on several axes; entries are drawn directly from the cited publications and are not re-derived or estimated by the present authors.

This highlights a scope distinction: [1], [3] report bit-cell/array-level leakage-stability gains via custom 8T transistor sizing – a circuit-level technique orthogonal to, and combinable with, the RTL-level gating proposed here. [5], [6] target compute-in-memory macros where power is dominated by analog/mixed-signal compute rather than a digital controller. [2] addresses the same architectural layer but reports trade-offs qualitatively rather than as post-synthesis figures, precluding direct numerical comparison. Within its own design point, this controller's 0.03% clock-network power share is consistent with the order-of-magnitude gating-based reductions reported in [6], [7], though a like-for-like (same node/array/workload) benchmark against a re-implemented baseline was not performed and is the most direct way to strengthen this comparison.

TABLE IV
COMPARISON WITH REPRESENTATIVE LOW-POWER SRAM/MEMORY-CONTROLLER DESIGNS

Design	Node	Level	Reported Metric
This work	45 nm	Controller (RTL)	9.68 mW total @ 100 MHz, 256×64 array
Chang *et al.* [3]	65 nm PD-SOI	Custom bit-cell	32 kb array: 5.3 GHz @ 1.2 V; 295 MHz @ 0.41 V
Verma & Chandrakasan [1]	65 nm	Custom bit-cell	Sub-threshold standby leakage reduction (cell-level)
Le Gallo *et al.* [5]	Mixed-signal	CIM chip	Inference-engine level, not directly comparable
Ren *et al.* [6]	Sub-100 nm	CIM macro	Reported at macro/inference level
Goossens *et al.* [2]	N/A	Controller (arch.)	Architectural; no post-synthesis figure

G. Latency, Throughput, and Energy Trade-offs Under Varying Workloads

The two-tier gating strategy (Section II-D) exposes a latency–energy trade-off governed by workload duty cycle. Under *high-activity* workloads (*cs* asserted most cycles), *mem_enable* is nearly always active, fine-grained gating saves little energy, and throughput is limited only by the 100 MHz clock and the 8.14 ns critical-path latency (Section IV-C) – one access per cycle, no wake-up penalty. Under *low-activity/bursty* workloads, entering *sleep* between bursts removes essentially all dynamic and clock-network power (Table II) but reintroduces wake-up latency: *pwr_gate* must release, the rail settle (*pwr_good* re-assert), and only then *clk_gated* resume, before the next access. This latency depends on the header/footer switch and supply-network time constant – physical-design parameters outside this RTL/synthesis scope – so it is described qualitatively rather than quantified without supporting extraction data. The *idle* signal, available every cycle at negligible cost (Section IV-A), lets a system integrator implement a workload-adaptive policy (e.g., asserting *sleep* only after *idle* persists beyond the wake-up latency) so power-gating benefit is realized only where not offset by wake-up cost – tuning that threshold against a measured wake-up latency is natural future work.

V. CONCLUSION

This paper presents the design, simulation, and synthesis of a low-power SRAM controller in 45 nm CMOS. At the RTL level, *clk_gated* and *pwr_gate*, interlocked with the datapath

through *pwr_good*, eliminate unnecessary dynamic/leakage power during idle periods while guarding against reads/writes on an unstable supply. Post-synthesis results show 26,297 cells and 9.68 mW at 100 MHz, with power-management logic contributing $\sim$0.03% of cell count. The +1658 ps slack ($\sim$17% of period) provides margin against expected post-layout degradation and PVT variation, though full parasitic-extracted post-layout STA remains future work. Functional verification confirmed correct sequencing across initialization, access, sleep, and wake-up for directed scenarios; broader constrained-random and hardware/FPGA validation are open items. CTS and layout results validate balanced clock distribution and clean routing. Next steps: (i) full-corner PVT sign-off and standby-leakage characterization, (ii) parasitic-extraction-based post-layout verification, (iii) constrained-random/FPGA validation, and (iv) adaptive voltage scaling and multi-V_{th} assignment to further improve the power–delay trade-off.

ACKNOWLEDGMENT

The authors acknowledge the Visvesvaraya PhD Scheme, MeitY, Govt. of India and the support of the Chips to Startup (C2S) Programme, Ministry of Electronics and Information Technology (MeitY), Government of India, through the ChipIN initiative for providing the EDA tools and technology node access necessary for this research.

REFERENCES

[1] N. Verma and A. P. Chandrakasan, "A 256 kb 65 nm 8T subthreshold SRAM employing sense-amplifier redundancy," *IEEE J. Solid-State Circuits*, vol. 43, no. 1, pp. 141–149, Jan. 2008. doi: 10.1109/jssc.2007.908005

[2] S. S. Goossens, J. J. Kuijsten, K. B. Akesson, and K. K. Goossens, "A reconfigurable real-time SDRAM controller for mixed time-criticality systems," TU/E Research Portal, Eindhoven Univ. of Technology, 2013. [Online]. Available: http://library.tue.nl/csp/dare/LinkToRepository.csp?recordnumber=770699

[3] L. Chang, R. K. Montoye, Y. Nakamura, K. A. Batson, R. J. Eickemeyer, R. H. Dennard, W. Haensch, and D. Jamsek, "An 8T-SRAM for variability tolerance and low-voltage operation in high-performance caches," *IEEE J. Solid-State Circuits*, vol. 43, no. 4, pp. 956–963, Apr. 2008. doi: 10.1109/jssc.2007.917509

[4] A. Baz, D. Shang, F. Xia, and A. Yakovlev, "Self-timed SRAM for energy harvesting systems," in *Lecture Notes in Computer Science*, pp. 105–115, 2011. doi: 10.1007/978-3-642-17752-1_11

[5] M. L. Gallo et al., "A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference," *Nature Electronics*, vol. 6, no. 9, pp. 680–693, Sep. 2023. doi: 10.1038/s41928-023-01010-1

[6] E. Ren, J. Liu, L. Luo, X. Yang, Q. Wei, and F. Qiao, "NS-Engine: Near-sensor neural network engine with SRAM-based compute-in-memory macro," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, pp. 1–5, 2024. doi: 10.1109/iscas58744.2024.10557977

[7] G. Kalokerinos, V. Papaefstathiou, G. Nikiforos, S. Kavadias, M. Katevenis, D. Pnevmatikatos, and X. Yang, "FPGA implementation of a configurable cache/scratchpad memory with virtualized user-level RDMA capability," in *Proc. IEEE Int. Conf. Embedded Comput. Syst.: Archit., Model. Simul. (ICSAMOS)*, pp. 149–156, 2009. doi: 10.1109/icsamos.2009.5289226

[8] S. K. Kingra, V. Parmar, C. Chang, B. Hudec, T. Hou, and M. Suri, "SLIM: Simultaneous logic-in-memory computing exploiting bilayer analog OXRAM devices," *Scientific Reports*, vol. 10, no. 1, p. 2567, Feb. 2020. doi: 10.1038/s41598-020-59121-0

[9] J. C. Knight and T. Nowotny, "GPUs outperform current HPC and neuromorphic solutions in terms of speed and energy when simulating a highly-connected cortical model," *Frontiers in Neuroscience*, vol. 12, p. 941, Dec. 2018. doi: 10.3389/fnins.2018.00941

[10] J. Zhao, C. Xu, P. Chi, and Y. Xie, "Memory and storage system design with nonvolatile memory technologies," *IPSJ Trans. Syst. LSI Design Methodology*, vol. 8, pp. 2–11, 2015. doi: 10.2197/ipsjtsldm.8.2

[11] S. K. Kingra, V. Parmar, C. Chang, B. Hudec, T. Hou, and M. Suri, "SLIM: Simultaneous logic-in-memory computing exploiting bilayer analog OXRAM devices," *Scientific Reports*, vol. 10, no. 1, p. 2567, 2020. doi: 10.1038/s41598-020-59121-0

[12] M. L. Gallo et al., "A 64-core mixed-signal in-memory compute chip based on phase-change memory for deep neural network inference," *Nature Electronics*, vol. 6, no. 9, pp. 680–693, 2023. doi: 10.1038/s41928-023-01010-1